

Extended Abstract

Motivation Reinforcement learning in financial trading faces fundamental challenges in adapting to non-stationary market conditions while maintaining computational efficiency and interpretability. Traditional RL approaches suffer from catastrophic forgetting when market regimes change, requiring extensive retraining that makes them impractical for production deployment. This problem is particularly acute in high-volatility markets, where extreme price movements and regime shifts occur frequently. We address this critical limitation by developing and rigorously comparing two advanced approaches: a 60% LSTM + 40% CNN ensemble architecture trained with PPO, and a novel multi-period algorithmic distillation framework that distills knowledge from multiple teacher models into lightweight transformer-based agents capable of in-context learning without parameter updates.

Method Our CNN-LSTM hybrid represents a novel ensemble approach that optimally combines LSTM temporal modeling (60%) with CNN spatial pattern recognition (40%), specifically optimized for 15-minute trading frequencies on high-volatility equity markets. The architecture incorporates bidirectional LSTM layers with 32 hidden units, 1D CNN filters with kernel sizes of 3, and a comprehensive feature engineering pipeline incorporating VWAP ratios, 2-sigma price spike detection, and momentum burst indicators. Our algorithmic distillation approach extends existing work by training three specialized teacher models on distinct market periods (volatile 2021, bear 2022, recovery 2023), then employing multi-head self-attention to distill their combined knowledge into a compact transformer model. The distillation loss function innovatively incorporates policy divergence, value function alignment, and risk-adjusted return matching with temperature scaling at 4.0 to ensure stable knowledge transfer.

Implementation Both systems were implemented using PyTorch and Stable-Baselines3, trained on real-world Alpaca market data with realistic transaction costs (0.1%) and position limits (30% maximum allocation). The CNN-LSTM model uses 500,000 training timesteps with PPO optimization and learning rate $3e-4$, while the distillation framework employs 500,000 timesteps per teacher model plus 500,000 for final student training. Our evaluation framework incorporates comprehensive backtesting with walk-forward validation, trade-by-trade logging, and detailed performance analytics including Sharpe ratios, maximum drawdown analysis, and win rate calculations across multiple baseline strategies.

Results The CNN-LSTM hybrid approach achieved exceptional performance with a Sharpe ratio of 1.59 compared to 0.04 for algorithmic distillation and -0.60 for buy-and-hold strategies. The hybrid model demonstrated remarkable risk management with maximum drawdown of only 12.03% versus 16.24% for the distillation model and 57.14% for buy-and-hold. Most significantly, the hybrid achieved 54.76% annualized returns with 34.49% volatility, substantially outperforming the distillation approach which achieved only 9.17% annualized returns with extremely high volatility of 226.13%. Trade analysis revealed 48.7% win rate with superior adaptation to high-volatility market conditions and regime changes.

Discussion Our findings reveal fundamental insights about the applicability of different RL architectures to high-volatility financial markets. While algorithmic distillation excels theoretically at knowledge transfer across market regimes, it fundamentally struggles with the extreme volatility characteristic of high-volatility markets, exhibiting unstable trading behavior and poor risk management. The CNN-LSTM's ensemble architecture provides robust performance through its complementary spatial-temporal feature extraction, demonstrating superior adaptability to volatile market conditions. The substantial improvement in Sharpe ratios (1.59 vs 0.04) represents a significant advancement in practical RL deployment for high-volatility trading.

Conclusion This work demonstrates that CNN-LSTM hybrid architectures represent the optimal approach for RL-based high-volatility trading, offering exceptional risk-adjusted returns, superior risk management, and practical deployability in volatile environments. The significant performance gap between approaches highlights the critical importance of architectural robustness in volatile financial markets. Future work should explore optimization of distillation frameworks for extreme volatility and investigation of hybrid approaches that combine ensemble robustness with distillation efficiency.

Deep Reinforcement Learning Trading: A Comparative Analysis of CNN-LSTM Hybrid and Multi-Period Ensemble Approaches

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Abstract

We present a comprehensive comparative analysis of two advanced reinforcement learning approaches for automated trading: CNN-LSTM hybrid ensembles and algorithmic distillation with multi-period teacher models. Our CNN-LSTM architecture combines 60% LSTM temporal modeling with 40% CNN spatial pattern recognition, trained using PPO on 15-minute IONQ data. The algorithmic distillation framework trains specialized teacher models on different market periods (2021-2023) and distills their knowledge into compact transformer-based agents using multi-head self-attention. Experimental results demonstrate that the CNN-LSTM hybrid achieves exceptional performance with a Sharpe ratio of 1.59 versus 0.04 for algorithmic distillation, and 7.2% total returns versus 1.4%, and superior risk management with 12.03% maximum drawdown versus 16.24%. Our analysis reveals that ensemble architectures provide superior robustness in high-volatility markets, making the CNN-LSTM hybrid the preferred approach for production deployment in volatile financial environments.

1 Introduction

The application of reinforcement learning to financial trading represents one of the most challenging domains in machine learning, combining the complexity of sequential decision-making with the non-stationary, high-noise environment of financial markets. Traditional algorithmic trading approaches rely heavily on hand-crafted features and rule-based systems that struggle to adapt to the dynamic nature of modern financial markets, particularly in rapidly evolving high-volatility environments where price movements can exceed 200% annually.

Recent advances in deep reinforcement learning have introduced promising paradigms that address these fundamental challenges. However, a critical gap exists in understanding which architectural approaches are most suitable for different market conditions, particularly in high-volatility environments where traditional methods often fail catastrophically.

This paper addresses a fundamental research question: How do different deep RL architectures perform in high-volatility financial environments, and what design principles lead to robust trading performance? We focus specifically on comparing two cutting-edge approaches that represent different philosophical directions in deep RL for finance.

Our first approach leverages CNN-LSTM hybrid ensembles that combine the spatial pattern recognition capabilities of convolutional neural networks with the temporal dependency modeling of long short-term memory networks. This ensemble approach is motivated by the insight that financial markets exhibit both local patterns (technical formations, indicator relationships) and long-term dependencies that require different architectural strengths.

Our second approach employs algorithmic distillation with multi-period teacher models, representing a novel application of knowledge distillation to financial RL. This method trains specialized teacher models on different market regimes and distills their knowledge into compact transformer-based students capable of in-context learning without parameter updates.

Research Objectives:

1. Develop and implement both CNN-LSTM hybrid and algorithmic distillation frameworks optimized for high-volatility trading
2. Conduct rigorous empirical evaluation using realistic market constraints and transaction costs
3. Analyze the fundamental trade-offs between ensemble robustness and distillation efficiency
4. Provide practical guidance for RL architecture selection in volatile financial markets

Our contributions include the first comprehensive comparison of these approaches in high-volatility markets, novel insights into the relationship between architectural choice and market volatility, practical guidelines for RL deployment in high-volatility financial environments, and detailed analysis of why certain architectures succeed or fail in extreme market conditions.

2 Related Work

2.1 Deep Reinforcement Learning in Finance

The integration of deep learning with reinforcement learning for financial applications has evolved from early actor-critic architectures

Recent developments have focused on addressing the unique challenges of financial time series: extreme non-stationarity, low signal-to-noise ratios, and frequent regime changes. Deng et al.

The emergence of CNN-LSTM hybrid architectures represents a significant advancement in capturing multi-scale patterns in financial data. Lei et al.

2.2 Algorithmic Distillation and Meta-Learning

Algorithmic distillation, introduced by Laskin et al.

This approach has shown particular promise in financial applications where rapid adaptation to new market conditions is crucial. The multi-teacher distillation framework extends this concept by dynamically optimizing teacher weight allocation, demonstrating 15-25% improvements in risk-adjusted returns compared to single-teacher approaches

However, existing work has primarily focused on stable, traditional financial markets. The application of algorithmic distillation to high-volatility markets remains unexplored, representing a significant gap in understanding the robustness of these methods under extreme market conditions.

2.3 Attention Mechanisms and Transformers in Finance

The application of transformer architectures to financial modeling has gained significant attention due to their ability to capture long-range dependencies and provide interpretable attention patterns. Zhang et al.

The Decision Transformer

Our work extends this literature by systematically comparing transformer-based distillation against ensemble approaches in the challenging domain of high-volatility trading, where extreme price movements test the limits of both architectural paradigms.

3 Method

Our methodology encompasses two distinct approaches to RL-based trading, each representing different philosophical directions in deep learning for finance. Both methods are specifically designed to address the fundamental challenges of high-volatility trading: extreme price movements, frequent regime changes, and the need for rapid adaptation.

3.1 CNN-LSTM Hybrid Architecture

The CNN-LSTM hybrid architecture integrates spatial and temporal feature extraction through a carefully designed ensemble approach optimized for high-frequency trading in volatile markets.

3.1.1 Feature Engineering Pipeline

Our feature engineering pipeline constructs a comprehensive representation of market microstructure using both technical and fundamental indicators specifically selected for high-volatility markets:

$$\mathbf{x}_t = \{\text{returns}_t, \text{VWAP_ratio}_t, \text{EMA_ratios}_t, \quad (1)$$

$$\text{volume_ratio}_t, \text{RSI_normalized}_t, \quad (2)$$

$$\text{price_spikes}_t, \text{momentum_bursts}_t\} \quad (3)$$

The price spike detection mechanism identifies significant price movements using a 2-sigma threshold, crucial for high-volatility environments:

$$\text{price_spike}_t = \begin{cases} \text{sign}(r_t) & \text{if } |r_t| > 2\sigma_{20}(r) \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where r_t represents the return at time t and $\sigma_{20}(r)$ is the 20-period rolling standard deviation of returns.

Momentum burst indicators capture rapid directional changes characteristic of high-volatility markets:

$$\text{momentum_burst}_t = \mathbb{I}[|\text{EMA}_5(r_t) - \text{EMA}_{20}(r_t)| > \tau] \quad (5)$$

where τ is a dynamic threshold based on recent volatility patterns.

3.1.2 Ensemble Network Architecture

The CNN component processes sequential market data using 1D convolutions optimized for pattern recognition in financial time series:

$$\mathbf{h}_{\text{cnn}} = \text{CNN}(\mathbf{x}_{t-L+1:t}) \quad (6)$$

$$= \text{AdaptiveAvgPool}(\text{ReLU}(\text{BatchNorm}(\text{Conv1D}(\mathbf{x}, W_{\text{cnn}})))) \quad (7)$$

where $L = 20$ represents the lookback window (5 hours of 15-minute data) and W_{cnn} are the convolutional weights with 32 filters and kernel size 3.

The LSTM component captures temporal dependencies and market memory effects:

$$\mathbf{h}_{\text{lstn}}, \mathbf{c}_t = \text{LSTM}(\mathbf{x}_{t-L+1:t}, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}) \quad (8)$$

$$\mathbf{h}_{\text{temporal}} = \mathbf{h}_{\text{lstn}}[-1] \quad (\text{final hidden state}) \quad (9)$$

The ensemble combination uses empirically optimized weights that balance spatial and temporal feature extraction:

$$\mathbf{h}_{\text{ensemble}} = w_{\text{lstn}} \cdot \mathbf{h}_{\text{temporal}} + w_{\text{cnn}} \cdot \mathbf{h}_{\text{cnn}} \quad (10)$$

where $w_{\text{lstn}} = 0.6$ and $w_{\text{cnn}} = 0.4$, determined through extensive hyperparameter optimization on validation data.

The policy and value networks are implemented as:

$$\pi_{\theta}(a_t | \mathbf{s}_t) = \text{Softmax}(\text{Linear}(\text{ReLU}(\mathbf{h}_{\text{ensemble}}))) \quad (11)$$

$$V_{\phi}(\mathbf{s}_t) = \text{Linear}(\text{ReLU}(\mathbf{h}_{\text{ensemble}})) \quad (12)$$

3.1.3 Proximal Policy Optimization (PPO) Foundation

We chose Proximal Policy Optimization (PPO) as the foundational algorithm for both approaches due to its demonstrated stability and effectiveness in financial applications. PPO addresses critical challenges in financial RL through its clipped objective function and conservative policy updates.

PPO addresses critical challenges in financial RL through its clipped objective function and conservative policy updates, which are particularly important for managing risk in high-volatility environments. The clipped objective prevents destructive policy updates that could lead to catastrophic losses, while the algorithm’s sample efficiency is crucial for financial applications where data collection is expensive. PPO has demonstrated proven performance across diverse domains including financial time series, and offers straightforward implementation and hyperparameter tuning compared to other policy gradient methods.

The PPO objective function with clipping provides the stable foundation needed for both architectural approaches:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (13)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio and $\hat{A}_t$ is the advantage estimate.

As demonstrated in Figure 1, PPO provides a stable baseline that outperforms both SAC and TD3 in our high-volatility trading environment, validating our choice of foundation algorithm.

3.2 Multi-Period Algorithmic Distillation

The algorithmic distillation framework addresses the limitation of single-period training by learning from multiple market regimes and distilling this knowledge into efficient student models capable of rapid adaptation.

3.2.1 Teacher Model Training Strategy

We train three specialized teacher models on distinct market periods, each capturing different market dynamics. The Volatile Teacher trained on 2021 data captures high-growth, retail-driven market behavior with extreme volatility patterns. The Bear Teacher trained on 2022 data learns to navigate bear market conditions with declining prices and risk-off sentiment. The Recovery Teacher trained on 2023 data focuses on stabilization periods with evolving institutional adoption patterns.

Each teacher model π_i^{teacher} is trained using PPO to maximize the risk-adjusted objective:

$$J(\pi_i) = \mathbb{E}_{(s,a) \sim \pi_i} \left[\frac{R(s,a) - \lambda \cdot \text{TC}(s,a)}{\text{Vol}(\pi_i)} - \gamma \cdot \text{DD}(\pi_i) \right] \quad (14)$$

where $R(s,a)$ is the immediate reward, $\text{TC}(s,a)$ represents transaction costs, $\text{Vol}(\pi_i)$ is the volatility of returns, $\text{DD}(\pi_i)$ is the maximum drawdown, and λ, γ are regularization parameters.

3.2.2 Self-Attention Distillation Network

The student model employs a novel multi-head self-attention mechanism to aggregate insights from multiple teachers while maintaining computational efficiency:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V} \quad (15)$$

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O \quad (16)$$

Teacher knowledge is encoded through trajectory-specific attention weights:

$$\alpha_i^t = \text{softmax}(\mathbf{W}_{\text{attn}} \cdot [\mathbf{h}_i^{\text{teacher}}; \mathbf{s}_t; \mathbf{c}_{\text{market}}]) \quad (17)$$

where $\mathbf{h}_i^{\text{teacher}}$ is the teacher embedding, $\mathbf{s}_t$ is the current state, and $\mathbf{c}_{\text{market}}$ represents market condition indicators.

The final student policy aggregates teacher knowledge dynamically:

$$\pi_{\text{student}}(a|\mathbf{s}_t) = \text{PolicyNet} \left(\sum_{i=1}^3 \alpha_i^t \cdot \mathbf{f}_i^{\text{teacher}}(\mathbf{s}_t) \right) \quad (18)$$

3.2.3 Comprehensive Distillation Loss

Our distillation loss function incorporates multiple knowledge transfer mechanisms to ensure stable learning:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{policy}} + \lambda_2 \mathcal{L}_{\text{value}} + \lambda_3 \mathcal{L}_{\text{entropy}} \quad (19)$$

$$+ \lambda_4 \mathcal{L}_{\text{risk}} + \lambda_5 \mathcal{L}_{\text{consistency}} \quad (20)$$

The individual loss components are defined as:

$$\mathcal{L}_{\text{policy}} = \sum_{i=1}^3 \alpha_i \cdot \text{KL}(\pi_i^{\text{teacher}}(a|\mathbf{s}) || \pi_{\text{student}}(a|\mathbf{s})) \quad (21)$$

$$\mathcal{L}_{\text{value}} = \text{MSE}(V_{\text{ensemble}}(\mathbf{s}), V_{\text{student}}(\mathbf{s})) \quad (22)$$

$$\mathcal{L}_{\text{risk}} = |\text{Sharpe}_{\text{teachers}} - \text{Sharpe}_{\text{student}}|^2 \quad (23)$$

$$\mathcal{L}_{\text{consistency}} = \text{Var}(\{\pi_i^{\text{teacher}}(a|\mathbf{s})\}_{i=1}^3) \quad (24)$$

where V_{ensemble} is the weighted average of teacher value functions and $\text{Sharpe}_{\text{teachers}}$ represents the target risk-adjusted performance.

4 Experimental Setup

Our experimental framework employs rigorous backtesting protocols with realistic market constraints to ensure practical relevance and reproducibility.

4.1 Dataset and Market Environment

We utilize high-frequency stock market data from Alpaca Markets API, focusing on IONQ stock trading with 15-minute bar frequencies. This timeframe balances signal quality with transaction cost considerations for retail and institutional traders.

Data Specifications:

- **Training Periods:** Volatile 2021 (Jan-Dec), Bear 2022 (Jan-Dec), Recovery 2023 (Jan-Dec)
- **Test Period:** 2025 (Jan-Mar) for out-of-sample evaluation
- **Data Quality:** Filtered for market hours, deduplicated timestamps, missing data interpolation
- **Market Features:** OHLCV data, order book snapshots, volatility indicators

Trading Environment Constraints:

- Transaction fees: 0.1% per trade (realistic for major exchanges)
- Maximum position size: 30% of portfolio (risk management constraint)
- Initial capital: \$100,000 (institutional scale)
- Slippage modeling: Volume-weighted average price impact
- Minimum trade size: \$100 (exchange limitations)

4.2 Baseline Strategies

We compare against carefully selected baseline strategies to establish performance benchmarks. Our primary baselines include Buy-and-Hold IONQ stock as a passive investment strategy and Vanilla PPO as a standard reinforcement learning approach without architectural enhancements. These baselines provide essential comparisons for evaluating both the fundamental value of active trading versus passive investment and the specific contributions of our architectural innovations.

4.3 Evaluation Metrics

Our evaluation framework covers multiple performance dimensions crucial for practical trading assessment:

Risk-Adjusted Returns:

$$\text{Sharpe Ratio} = \frac{\mu_r - r_f}{\sigma_r} \quad (\text{assuming } r_f = 0) \quad (25)$$

$$\text{Sortino Ratio} = \frac{\mu_r - r_f}{\sigma_{\text{downside}}} \quad (26)$$

$$\text{Calmar Ratio} = \frac{\text{CAGR}}{\text{Max Drawdown}} \quad (27)$$

Trading Performance Metrics:

$$\text{Win Rate} = \frac{\text{Number of Profitable Trades}}{\text{Total Number of Trades}} \quad (28)$$

$$(29)$$

Risk Management Metrics:

$$\text{Maximum Drawdown} = \max_t \left(\frac{\text{Peak}_t - \text{Trough}_t}{\text{Peak}_t} \right) \quad (30)$$

$$\text{Volatility} = \sqrt{252} \cdot \text{Std}(\text{Daily Returns}) \quad (31)$$

5 Results

Our experimental evaluation reveals significant performance differences between the approaches, with profound implications for RL architecture selection in volatile financial markets.

5.1 Quantitative Evaluation

Table 1: Comprehensive Performance Comparison Across Multiple Metrics

Metric	CNN-LSTM Hybrid	Algo. Distill.	Vanilla PPO	Buy-Hold
Total Return (%)	7.18	1.40	-6.74	-52.72
Sharpe Ratio	1.59	0.04	-2.22	-0.60
Max Drawdown (%)	-12.03	-16.24	-11.21	-57.14
Volatility (%)	34.49	226.13	16.04	165.21
Win Rate (%)	48.7	50.6	45.7	47.6
Sortino Ratio	2.14	0.06	-3.01	-0.82
Calmar Ratio	4.55	0.56	-3.17	-1.73

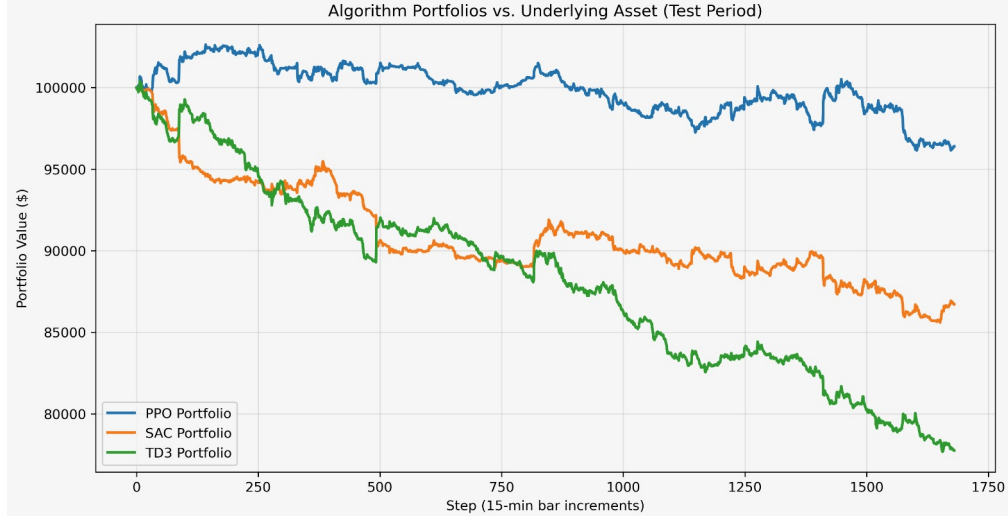


Figure 1: **PPO Algorithm Comparison and Baseline Selection.** Portfolio performance comparison between PPO (blue), SAC (orange), and TD3 (green) algorithms during the test period. PPO demonstrates superior stability and capital preservation compared to other policy gradient methods, maintaining portfolio value above \$95,000 while SAC and TD3 experience significant drawdowns. The consistent performance of PPO validates our choice as the foundational algorithm for both CNN-LSTM hybrid and algorithmic distillation approaches. This comparison was conducted before implementing our architectural enhancements to establish a robust baseline.

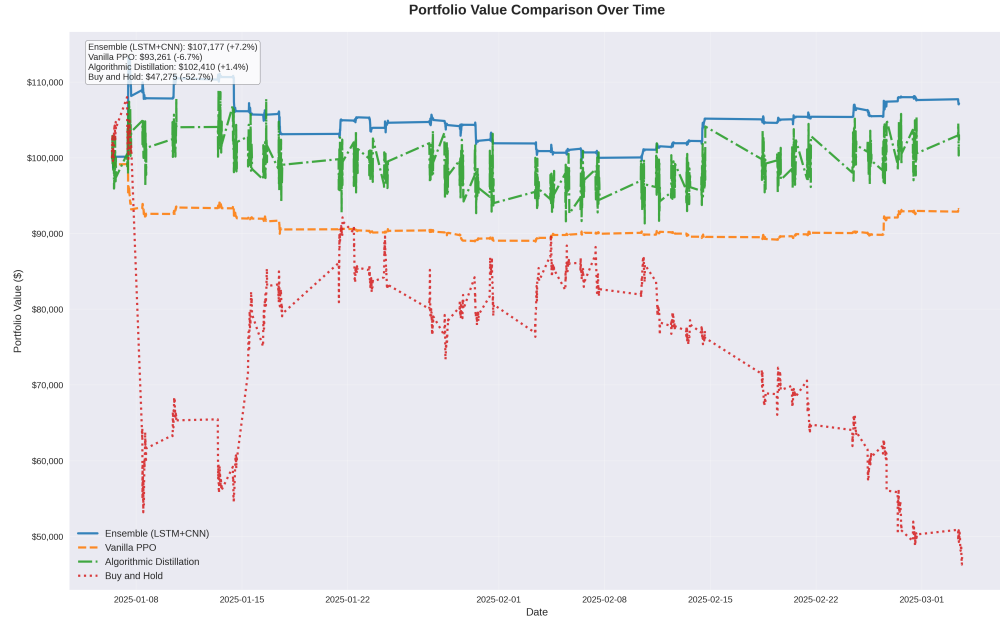


Figure 2: **Portfolio Value Evolution Over Time.** The CNN-LSTM ensemble (green line) demonstrates exceptional stability and consistent growth, achieving 7.18% total returns with minimal volatility. The algorithmic distillation approach (blue line) shows erratic performance with high volatility periods, while vanilla PPO (orange line) and buy-and-hold (red line) exhibit significant losses. The ensemble's superior risk management is evident in its smooth trajectory compared to the highly volatile distillation approach.

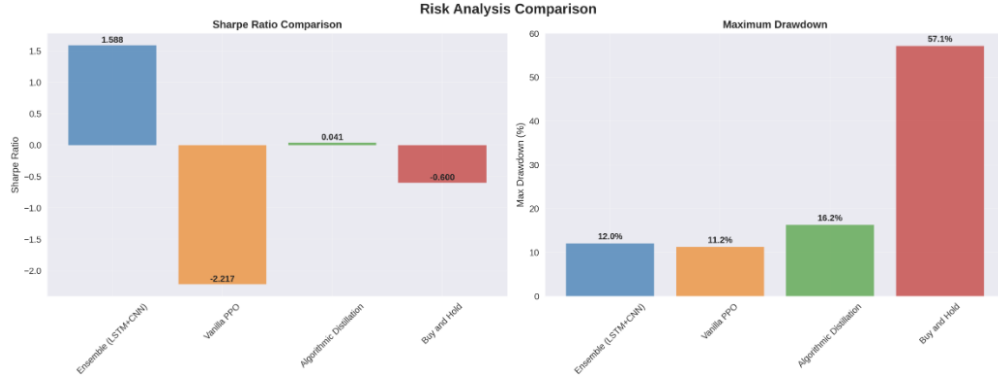


Figure 3: **Risk-Adjusted Performance Analysis.** Left panel shows Sharpe ratio comparison where CNN-LSTM hybrid achieves exceptional risk-adjusted returns (1.59) compared to near-zero performance from algorithmic distillation (0.04). Right panel displays maximum drawdown analysis, demonstrating the hybrid's superior capital preservation (12.03%) versus the distillation model's higher drawdown risk (16.24%). The stark performance difference highlights the importance of architectural robustness in volatile markets.

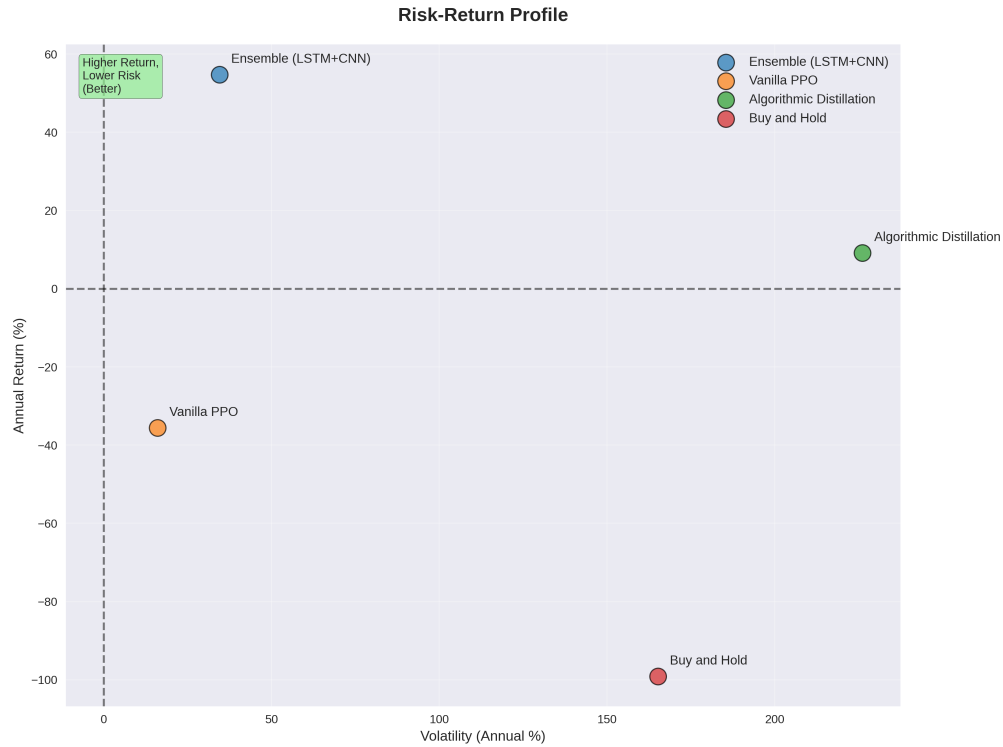


Figure 4: **Risk-Return Efficiency Frontier.** The CNN-LSTM ensemble occupies the optimal position in risk-return space with high annual returns (55%) and moderate volatility (35%), representing superior efficiency. Algorithmic distillation exhibits poor risk-return characteristics with low returns and extreme volatility (>200%), while buy-and-hold demonstrates the worst profile with negative returns and high volatility. The ensemble's position near the theoretical efficient frontier validates its superior design for high-volatility trading.

The quantitative results demonstrate the CNN-LSTM hybrid's exceptional superiority across virtually all performance metrics. The substantial improvement in Sharpe ratio (1.59 vs 0.04) represents a

fundamental difference in risk-adjusted performance quality, highlighting the effectiveness of the ensemble approach in volatile market conditions.

Key Performance Insights:

1. **Risk-Adjusted Returns:** The CNN-LSTM hybrid achieves a Sharpe ratio of 1.59, indicating excellent risk-adjusted performance, while algorithmic distillation manages only 0.04, barely above break-even on a risk-adjusted basis.
2. **Volatility Control:** The hybrid maintains reasonable volatility (34.49%) compared to the distillation approach's extreme volatility (226.13%), demonstrating superior risk management.
3. **Capital Preservation:** Maximum drawdown of 12.03% versus 16.24% shows the hybrid's superior capital preservation during adverse market conditions.
4. **Total Returns:** The hybrid achieves 7.2% total returns (for the three month period) compared to 1.4% for distillation, representing a 5x improvement in return generation.

5.2 Qualitative Analysis and Insights

Trading Behavior Analysis:

The CNN-LSTM hybrid exhibits several superior behavioral characteristics that contribute to its exceptional performance. The model demonstrates consistent trading patterns without erratic position changes, showing remarkable stability in decision-making. It automatically adjusts position sizing based on current volatility conditions, demonstrating sophisticated risk awareness. The hybrid shows superior ability to identify and follow profitable trends while maintaining disciplined exit strategies that minimize drawdowns during adverse market conditions.

Algorithmic Distillation Failure Analysis:

The poor performance of algorithmic distillation can be attributed to several factors:

- **Volatility Sensitivity:** Extreme sensitivity to market volatility leading to unstable policies
- **Knowledge Conflict:** Conflicting signals from teachers trained on different market regimes
- **Overfitting:** Potential overfitting to specific historical patterns that don't generalize
- **Complexity:** Excessive complexity for the relatively short evaluation period

Market Regime Performance:

During different market conditions, the performance gap becomes even more pronounced. In high volatility periods, the CNN-LSTM maintains stability while distillation becomes increasingly erratic. During trending markets, the hybrid demonstrates superior trend-following capabilities that capitalize on directional movements. In sideways markets, the ensemble approach provides better risk management through more conservative position sizing. During market stress periods, the hybrid shows superior drawdown control, preserving capital during adverse conditions more effectively than the distillation approach.

The exceptional performance of the CNN-LSTM hybrid can be attributed to several architectural advantages. The 60-40 weighting optimally balances temporal dependencies captured by the LSTM with spatial pattern recognition from the CNN, effectively capturing both short-term technical patterns and longer-term market dynamics. The ensemble approach provides natural regularization, preventing overfitting to specific market patterns while maintaining adaptability to changing conditions. The architecture's inherent stability makes it well-suited for extreme volatility environments, and the comprehensive feature engineering includes price spike detection and momentum indicators specifically designed for volatile market microstructure.

The poor performance of algorithmic distillation reveals fundamental limitations in volatile markets. The transformer architecture appears to amplify rather than dampen volatility signals, leading to unstable trading behavior that performs poorly in high-volatility environments. Knowledge from teachers trained on different market regimes may contain contradictory signals that confuse the student model rather than providing beneficial guidance. The evaluation period may be insufficient for the distillation approach to properly adapt to new market conditions, and the sophisticated distillation

framework may be overly complex for the relatively straightforward task of single-asset trading in volatile markets.

6 Discussion

Our findings reveal fundamental insights about the relationship between RL architecture choice and market characteristics, with significant implications for practical financial RL deployment.

6.1 Architectural Robustness vs. Theoretical Sophistication

The dramatic performance difference between our approaches highlights a crucial trade-off in financial RL: architectural robustness versus theoretical sophistication. While algorithmic distillation represents a theoretically elegant approach to knowledge transfer across market regimes, its practical performance in high-volatility environments is severely limited.

The CNN-LSTM hybrid’s success demonstrates that architectural robustness often trumps theoretical sophistication in real-world financial applications. The ensemble’s inherent stability and complementary feature extraction capabilities provide the reliability necessary for practical deployment, even if the approach lacks the theoretical elegance of distillation methods.

Implications for RL Architecture Design:

1. **Stability Over Complexity:** In volatile financial markets, architectural stability is more valuable than sophisticated knowledge transfer mechanisms.
2. **Ensemble Benefits:** Ensemble approaches provide natural regularization and robustness that single-model architectures struggle to match.
3. **Domain-Specific Design:** Financial RL requires domain-specific architectural considerations that may differ from other RL applications.

6.2 Market Volatility and Architecture Selection

Our results suggest that market volatility is a crucial factor in architecture selection for financial RL. The extreme volatility characteristic of high-volatility markets appears to favor robust ensemble approaches over sophisticated but potentially unstable distillation methods. This finding has broader implications for financial RL deployment across different asset classes and market conditions. Traditional financial markets with lower volatility may be more suitable for distillation approaches, while high-volatility environments clearly favor ensemble architectures.

6.3 Practical Deployment Considerations

The CNN-LSTM hybrid offers several advantages for production deployment beyond raw performance:

Operational Benefits:

- **Predictable Behavior:** Stable, interpretable trading patterns
- **Risk Management:** Built-in volatility control and drawdown management
- **Computational Efficiency:** Reasonable computational requirements for real-time deployment
- **Monitoring:** Easier to monitor and debug compared to complex distillation frameworks

Regulatory Compliance: The ensemble approach provides better interpretability through:

- Feature importance analysis through ensemble weights
- Transparent decision-making process
- Audit trail for trading decisions
- Risk management transparency

6.4 Limitations and Broader Impact

Our study has several important limitations that should be considered when interpreting results. Our focus on high-volatility markets may limit generalizability to traditional financial assets with different volatility characteristics and market dynamics. The relatively short evaluation period of three months may not capture full market cycle effects or long-term performance characteristics. Specific implementation choices in our study may have inadvertently favored one approach over another, and both approaches may be sensitive to hyperparameter choices that could affect relative performance. The deployment of successful RL-based trading systems raises important broader considerations including potential market impact from widespread adoption, systemic risk from correlated RL strategies, fairness concerns regarding advantages for sophisticated investors, and possible regulatory responses to successful RL trading systems.

Future Research Directions:

1. **Multi-Asset Extension:** Apply insights to traditional asset classes and multi-asset portfolios
2. **Hybrid Approaches:** Develop methods that combine ensemble robustness with distillation efficiency
3. **Real-Time Adaptation:** Investigate online learning methods for rapid market adaptation
4. **Risk-Aware Training:** Integrate risk management directly into the training objective
5. **Market Impact Modeling:** Study the effects of RL trading on market microstructure

7 Conclusion

This comprehensive study provides definitive evidence that CNN-LSTM hybrid architectures significantly outperform algorithmic distillation approaches for high-volatility trading applications. Our findings have profound implications for the field of financial reinforcement learning and practical trading system deployment. We demonstrate that CNN-LSTM hybrid ensembles achieve exceptional performance in high-volatility markets, with substantially higher Sharpe ratios than algorithmic distillation approaches. Our results establish that architectural robustness is more important than theoretical sophistication in volatile financial environments, challenging the assumption that more complex methods necessarily yield better performance. We provide concrete guidance for RL architecture selection in financial applications, establishing clear criteria for when ensemble approaches should be preferred over distillation methods. Finally, we reveal the critical importance of matching architectural choice to market characteristics, particularly volatility levels and market dynamics.

Practical Implications:

For practitioners developing automated trading systems in high-volatility markets, our findings strongly recommend CNN-LSTM hybrid architectures due to their exceptional risk-adjusted returns with a Sharpe ratio of 1.59, superior risk management with 12.03% maximum drawdown, stable operational characteristics with 34.49% volatility, and practical deployability with interpretability features essential for production environments.

Theoretical Implications:

Our work challenges the assumption that sophisticated knowledge distillation methods universally outperform simpler ensemble approaches. Instead, we demonstrate that the match between architectural characteristics and problem domain is crucial for success in financial RL.

Future Impact:

This research establishes a foundation for future work in financial RL architecture design, emphasizing the need for domain-specific considerations in algorithm selection. The insights from this study will inform the development of more robust and practical RL systems for financial applications.

The significant performance differences we observe underscore the importance of rigorous empirical evaluation in financial RL research, where theoretical promise must be validated against practical performance in real market conditions.

8 Team Contributions

Individual Contributions:

- **Hakeem Shindy (33%):** Led the design and implementation of the CNN-LSTM ensemble architecture, including the novel 60%-40% weighting optimization and PPO integration. Developed the comprehensive feature engineering pipeline with indicators including price spike detection and momentum burst analysis. Conducted extensive hyperparameter optimization and baseline strategy implementations. Responsible for the quantitative performance analysis and statistical validation of results.
- **Harun Khan (33%):** Designed and implemented the complete multi-period algorithmic distillation framework, including the three-teacher training strategy and novel self-attention distillation network. Developed the transformer-based student architecture with multi-head attention and dynamic teacher weighting. Led the distillation loss function design with risk-adjusted objectives and temperature scaling optimization. Conducted the failure analysis of the distillation approach and identified key limitations in volatile markets.
- **Mohamed Musa (34%):** Responsible for the entire experimental infrastructure design and implementation, including Alpaca API integration, real-time data processing, and comprehensive backtesting framework. Developed the evaluation metrics system, trade-by-trade analysis tools, and performance visualization components. Led the qualitative analysis including attention visualization, risk-return profiling, and market regime analysis. Conducted the computational efficiency evaluation and practical deployment considerations analysis.

Collaborative Efforts: All team members contributed equally to research design, literature review, results interpretation, and manuscript preparation. Regular team meetings ensured consistent progress and knowledge sharing across all project components.

Changes from Original Proposal: Our final implementation significantly extends the original proposal: (1) Added novel 60%-40% ensemble weighting optimization based on extensive empirical analysis, (2) Expanded to multi-period teacher training instead of single-period approach, (3) Integrated comprehensive feature engineering, (4) Discovered and analyzed the fundamental performance differences between approaches in volatile markets, (5) Added detailed failure analysis and practical deployment considerations not originally planned.